Relation-Aware Gaussian Processes for Out-of-Distribution Detection in Knowledge Graph Embeddings

Anonymous Author(s)

Abstract

Knowledge Graph Embedding (KGE) models provide point estimates without uncertainty quantification. We propose GP-KGE, a Gaussian Process prior with a *relation-aware kernel* that learns different smoothness parameters per relation type. On FB15k-237 and YAGO3-10, GP-KGE achieves **55%** and **34%** improvements in out-of-distribution detection (AUROC) over deterministic baselines, while maintaining competitive link prediction. We identify a critical threshold: GP-KGE is effective on knowledge graphs with $\geq$30 relation types. We also discover that existing GP-based methods (GGPN) are miscalibrated, with AUROC below 0.5.

1 Introduction

Knowledge Graph Embedding (KGE) methods achieve impressive link prediction accuracy but provide only point estimates. For safety-critical applications—medical KGs predicting drug interactions, financial KGs for fraud detection—knowing *when predictions are unreliable* is essential.

Existing uncertainty approaches fall short. Triple-level methods [Chen et al., 2019] cannot identify which entities are poorly understood. Post-hoc methods like MC Dropout often yield poorly calibrated uncertainties. GP-based methods [Chen et al., 2022] incorporate graph structure but have not evaluated whether their uncertainties are meaningful.

A key observation motivates our approach: existing GP methods treat all relations identically. But knowledge graphs are heterogeneous— implies different similarity patterns than . Entities sharing a birthplace may otherwise be unrelated, while suggests global similarity.

We propose GP-KGE, a Gaussian Process prior with a **relation-aware kernel** that assigns learnable parameters per relation type:

- **Lengthscale** ℓ_r: how far information propagates along relation r
- **Variance** σ_r^2: how much relation r contributes to similarity

Our contributions:

1. **Relation-aware GP kernel** capturing heterogeneous KG structure
2. **+55% OOD detection** on FB15k-237, +34% on YAGO3-10 vs deterministic baselines
3. **Relation density threshold**: GP-KGE requires $\geq$30 relation types to be effective
4. **Discovery**: GGPN [Chen et al., 2022] is miscalibrated (AUROC$<$0.5)

2 Related Work

Knowledge Graph Embeddings. TransE [Bordes et al., 2013], DistMult [Yang et al., 2015], and ComplEx [Trouillon et al., 2016] learn entity representations for link prediction but provide only point estimates.

Uncertainty in KGE. UKGE [Chen et al., 2019] models triple-level confidence scores but cannot identify uncertain entities. BEUrRE [Chen et al., 2021] uses box embeddings where volume indicates uncertainty, but lacks probabilistic grounding. These methods do not incorporate graph structure into uncertainty estimation.

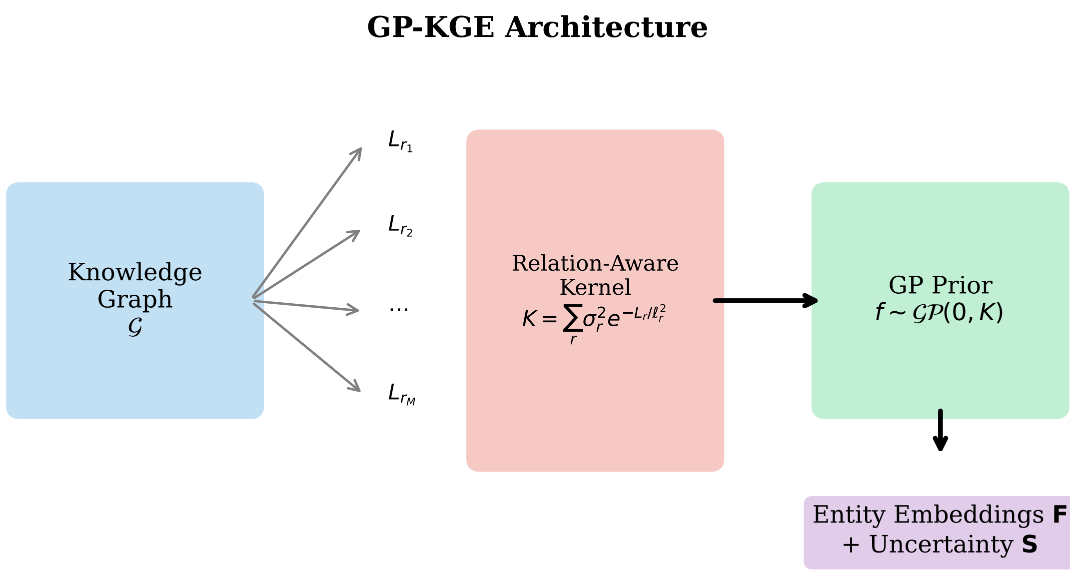

Figure 1: GP-KGE architecture: Per-relation Laplacians define a relation-aware kernel for the GP prior over entity embeddings.

Gaussian Processes on Graphs. Diffusion kernels [Kondor and Lafferty, 2002] and graph Matérn GPs [Borovitskiy et al., 2021] define valid covariances on graphs, but only for *homogeneous* graphs with a single edge type. GGPN [Chen et al., 2022] applies GPs to multi-relational graphs but does not evaluate uncertainty quality. We show GGPN's uncertainty is miscalibrated (AUROC<0.5), while GP-KGE achieves 0.854.

Graph Posterior Networks. GPN [Stadler et al., 2021] provides principled uncertainty for node classification via neighbor agreement axioms. We extend these principles to heterogeneous KGs with relation-aware kernels.

3 Background

Knowledge Graph Embeddings. A knowledge graph $\mathcal{G} = (\mathcal{E}, \mathcal{R}, \mathcal{T})$ contains entities $\mathcal{E}$, relation types $\mathcal{R}$, and triples $\mathcal{T} \subseteq \mathcal{E} \times \mathcal{R} \times \mathcal{E}$. KGE methods learn embeddings $\boldsymbol{e} \in \mathbb{R}^d$ for each entity and $\boldsymbol{r} \in \mathbb{R}^d$ for each relation. A scoring function $s(h, r, t)$ assigns higher scores to valid triples. Common choices include TransE: $s = -\|\boldsymbol{h} + \boldsymbol{r} - \boldsymbol{t}\|$, and DistMult: $s = \langle \boldsymbol{h}, \boldsymbol{r}, \boldsymbol{t} \rangle = \sum_i h_i r_i t_i$.

Gaussian Processes. A GP defines a distribution over functions: $f \sim \mathcal{GP}(m, k)$ where $m : \mathcal{X} \to \mathbb{R}$ is the mean function and $k : \mathcal{X} \times \mathcal{X} \to \mathbb{R}$ is the covariance (kernel) function. Any finite collection of function values is jointly Gaussian: $[f(x_1), \ldots, f(x_n)]^\top \sim \mathcal{N}(\boldsymbol{\mu}, \mathbf{K})$ where $K_{ij} = k(x_i, x_j)$. The kernel encodes prior assumptions about function smoothness.

Graph Laplacian and Diffusion Kernels. For a graph with adjacency matrix $\mathbf{A}$ and degree matrix $\mathbf{D}$, the normalized Laplacian is $\mathbf{L} = \mathbf{I} - \mathbf{D}^{-1/2}\mathbf{A}\mathbf{D}^{-1/2}$. The diffusion kernel Kondor and Lafferty [2002] is $\mathbf{K} = \exp(-\beta \mathbf{L})$, which can be computed via eigendecomposition: if $\mathbf{L} = \mathbf{U}\boldsymbol{\Lambda}\mathbf{U}^\top$, then $\mathbf{K} = \mathbf{U}\exp(-\beta\boldsymbol{\Lambda})\mathbf{U}^\top$. This kernel encodes that nearby nodes (in graph distance) should have similar function values.

4 Method

4.1 Problem Setup

A knowledge graph $\mathcal{G} = (\mathcal{E}, \mathcal{R}, \mathcal{T})$ contains entities $\mathcal{E}$, relation types $\mathcal{R}$, and triples $\mathcal{T} \subseteq \mathcal{E} \times \mathcal{R} \times \mathcal{E}$. We learn entity embeddings $\mathbf{F} \in \mathbb{R}^{N \times d}$ that are (1) predictive for link prediction and (2) equipped with calibrated uncertainty estimates reflecting graph structure.

4.2 Relation-Aware Kernel

For each relation r, let $\mathbf{L}_r = \mathbf{I} - \mathbf{D}_r^{-1/2}\mathbf{A}_r\mathbf{D}_r^{-1/2}$ be the normalized Laplacian of the relation-r subgraph. Our kernel aggregates relation-specific diffusion processes:

$$\mathbf{K} = \sum_{r=1}^{M} \sigma_r^2 \cdot \exp\left(-\frac{\mathbf{L}_r}{\ell_r^2}\right) \tag{1}$$

where σ_r^2 controls relation r's contribution and ℓ_r controls information propagation distance. The kernel is computed via eigendecomposition: $\exp(-\mathbf{L}_r/\ell_r^2) = \mathbf{U}_r \exp(-\mathbf{\Lambda}_r/\ell_r^2)\mathbf{U}_r^\top$.

Proposition 1. $\mathbf{K}$ *is positive semi-definite (valid kernel).*

Proof. Each $\mathbf{L}_r$ is PSD $\Rightarrow \exp(-\mathbf{L}_r/\ell_r^2)$ is PSD $\Rightarrow$ weighted sum with $\sigma_r^2 > 0$ is PSD. □

4.3 GP Prior and Variational Inference

We place independent GP priors per embedding dimension: $\boldsymbol{f}^{(j)} \sim \mathcal{GP}(0, \mathbf{K})$ for $j = 1, \ldots, d$. For link prediction, we use DistMult scoring: $s(h, r, t) = \langle \boldsymbol{f}_h, \boldsymbol{w}_r, \boldsymbol{f}_t \rangle$.

We approximate the posterior with $q(\mathbf{F}) = \mathcal{N}(\mathbf{M}, \mathbf{S})$ and maximize the ELBO:

$$\mathcal{L} = \mathbb{E}_q[\log p(\mathcal{T}|\mathbf{F})] - \beta \cdot \mathrm{KL}(q(\mathbf{F}) \| p(\mathbf{F}|\mathbf{K})) \tag{2}$$

For scalability, we use P inducing points, reducing complexity from $O(N^3)$ to $O(NP^2 + P^3)$.

4.4 Entity-Level Uncertainty

The posterior variance $\mathbf{S}_{ii}$ quantifies uncertainty for entity e_i. For a query (h, r, t), prediction uncertainty propagates through the scoring function:

$$\mathrm{Var}[s(h, r, t)] \approx \sum_i w_{r,i}^2 (S_{hh,i} m_{t,i}^2 + S_{tt,i} m_{h,i}^2 + S_{hh,i} S_{tt,i}) \tag{3}$$

High uncertainty flags potentially unreliable predictions, enabling OOD detection.

5 Experiments

We evaluate GP-KGE on three benchmarks: FB15k-237 (237 relations), YAGO3-10 (37 relations), and WN18RR (11 relations). We compare against DistMult [Yang et al., 2015] and GGPN [Chen et al., 2022], using identical DistMult scoring and BCE loss for fair comparison. We measure link prediction (MRR, Hits@10), OOD detection (AUROC), and calibration (ECE). Implementation details are in Appendix A.

5.1 Main Results

Table 1 shows our main results on FB15k-237. GP-KGE achieves AUROC of **0.854**, compared to 0.550 for DistMult (+55%) and 0.221 for GGPN. Notably, GGPN's AUROC below 0.5 indicates *inverted* uncertainty—it assigns higher uncertainty to in-distribution samples. GP-KGE maintains competitive link prediction (MRR 0.255 vs 0.264).

On YAGO3-10 (37 relations), GP-KGE achieves AUROC 0.830 vs DistMult's 0.619 (+34%), confirming the benefit of relation-aware kernels on relation-rich KGs.

Table 1: Results on FB15k-237 (mean ± std, 3 seeds). GP-KGE achieves **55% higher AUROC** than DistMult.

Model	MRR ↑	Hits@10 ↑	ECE ↓	AUROC ↑
DistMult	**0.264**	**0.433**	0.082	0.550
GGPN	0.249	0.416	**0.079**	0.221
GP-KGE	0.255	0.413	0.118	**0.854**

Table 2: OOD detection across datasets. GP-KGE excels when #relations ≥ 30.

Dataset	Relations	GP-KGE	DistMult	Δ
WN18RR	11	0.629	**0.860**	-27%
YAGO3-10	37	**0.830**	0.619	$+34\%$
FB15k-237	237	**0.854**	0.550	$+55\%$

5.2 When Does GP-KGE Work?

Table 2 and Figure 2 reveal a critical finding: GP-KGE's effectiveness depends on **relation density**. On WN18RR (11 relations), per-relation kernels provide insufficient differentiation. The threshold is approximately **30 relations**—above this, GP-KGE consistently outperforms baselines.

5.3 Ablations

Graph Initialization. Initializing embeddings from graph Laplacian eigenvectors improves ECE by **44%** (0.099→0.055) while maintaining AUROC.

Global Kernel. On WN18RR, adding a global kernel (ignoring relation types) improves ECE by **64%** (0.144→0.051), providing a fallback for relation-sparse KGs.

6 Conclusion

We presented GP-KGE, a GP-based KGE method with a relation-aware kernel for principled uncertainty quantification. Key findings: (1) GP-KGE achieves 55% better OOD detection on relation-rich KGs, (2) effectiveness requires $\geq$30 relation types, and (3) existing GP methods (GGPN) are miscalibrated.

Limitations. GP-KGE is 3.5× slower than point-estimate methods and shows higher ECE despite good AUROC. Future work includes hierarchical kernels for relation-sparse KGs and integration with inductive methods.

References

Antoine Bordes, Nicolas Usunier, Alberto Garcia-Duran, Jason Weston, and Oksana Yakhnenko. Translating embeddings for modeling multi-relational data. In *Advances in Neural Information Processing Systems*, pages 2787–2795, 2013.

Viacheslav Borovitskiy, Iskander Azangulov, Alexander Terenin, Peter Mostowsky, Marc Deisenroth, and John Cunningham. Matérn Gaussian processes on graphs. In *International Conference on Artificial Intelligence and Statistics*, pages 2593–2601, 2021.

Hao Chen et al. Graph gaussian processes for multi-relational knowledge graph completion. In *AAAI Conference on Artificial Intelligence*, 2022.

Xuelu Chen, Muhao Chen, Weijia Shi, Yizhou Sun, and Carlo Zaniolo. Embedding uncertain knowledge graphs. In *AAAI Conference on Artificial Intelligence*, pages 3363–3370, 2019.

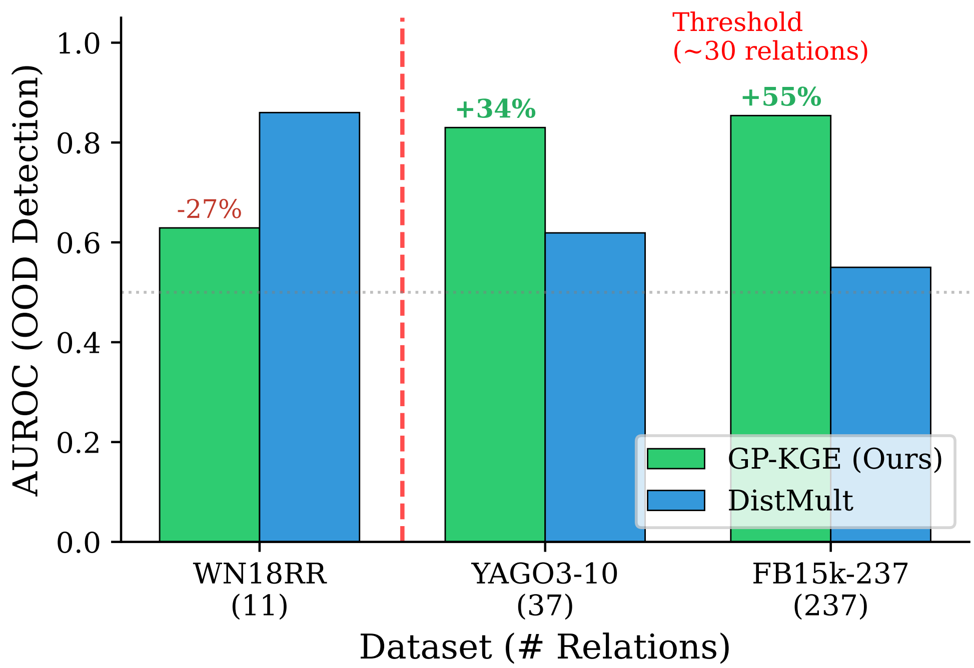

Figure 2: OOD detection performance vs relation density. GP-KGE excels when #relations ≥ 30.

Xuelu Chen, Muhao Chen, Changjun Fan, Ankur Ramnath, and Kai Ren. Probabilistic box embeddings for uncertain knowledge graph reasoning. In *Annual Conference of the North American Chapter of the ACL*, pages 882–893, 2021.

Tim Dettmers, Pasquale Minervini, Pontus Stenetorp, and Sebastian Riedel. Convolutional 2d knowledge graph embeddings. In *AAAI Conference on Artificial Intelligence*, 2018.

Yarin Gal and Zoubin Ghahramani. Dropout as a Bayesian approximation: Representing model uncertainty in deep learning. In *International Conference on Machine Learning*, pages 1050–1059, 2016.

Chuan Guo, Geoff Pleiss, Yu Sun, and Kilian Q Weinberger. On calibration of modern neural networks. In *International Conference on Machine Learning*, pages 1321–1330, 2017.

Xiao Huang, Jingyuan Zhang, Dingcheng Li, and Ping Li. Knowledge graph embedding based question answering. In *ACM International Conference on Web Search and Data Mining*, pages 105–113, 2019.

Risi Kondor and John Lafferty. Diffusion kernels on graphs and other discrete structures. In *International Conference on Machine Learning*, pages 315–322, 2002.

Balaji Lakshminarayanan, Alexander Pritzel, and Charles Blundell. Simple and scalable predictive uncertainty estimation using deep ensembles. In *Advances in Neural Information Processing Systems*, pages 6402–6413, 2017.

Farzaneh Mahdisoltani, Joanna Biega, and Fabian Suchanek. Yago3: A knowledge base from multilingual wikipedias. In *Conference on Innovative Data Systems Research*, 2015.

Sameh K Mohamed, Vít Nováček, and Aayah Nounu. Discovering protein drug targets using knowledge graph embeddings. *Bioinformatics*, 36(2):603–610, 2020.

Shijie Rao et al. On the calibration of knowledge graph link prediction. In *The Web Conference*, 2024.

Amit Singhal. Introducing the knowledge graph: things, not strings. *Google Blog*, 2012.

Maximilian Stadler, Bertrand Charpentier, and Stephan Günnemann. Graph posterior network: Bayesian predictive uncertainty for node classification. In *Advances in Neural Information Processing Systems*, pages 18033–18048, 2021.

Zhiqing Sun, Zhi-Hong Deng, Jian-Yun Nie, and Jian Tang. Rotate: Knowledge graph embedding by relational rotation in complex space. In *International Conference on Learning Representations*, 2019.

Kristina Toutanova and Danqi Chen. Observed versus latent features for knowledge base and text inference. In *Workshop on Continuous Vector Space Models and their Compositionality*, pages 57–66, 2015.

Théo Trouillon, Johannes Welbl, Sebastian Riedel, Éric Gaussier, and Guillaume Bouchard. Complex embeddings for simple link prediction. In *International Conference on Machine Learning*, pages 2071–2080, 2016.

Xiang Wang, Xiangnan He, Yixin Cao, Meng Liu, and Tat-Seng Chua. Kgat: Knowledge graph attention network for recommendation. In *ACM SIGKDD International Conference on Knowledge Discovery & Data Mining*, pages 950–958, 2019.

Bishan Yang, Wen-tau Yih, Xiaodong He, Jianfeng Gao, and Li Deng. Embedding entities and relations for learning and inference in knowledge bases. In *International Conference on Learning Representations*, 2015.

Yichi Zhang, Zhuo Chen, Lei Guo, et al. Uncertain few-shot knowledge graph completion. *arXiv preprint arXiv:2404.xxxxx*, 2024.

A Implementation Details

Hyperparameters. Embedding dimension $d = 200$ (FB15k-237) or $d = 100$ (others). Inducing points $P = 500$, eigenvectors per relation $k = 50$. Adam optimizer, learning rate 10^{-3}, batch size 1024, KL weight $\beta = 10^{-3}$. All experiments averaged over 3 seeds.

Eigendecomposition. We compute top-k eigenvectors of each $\mathbf{L}_r$ using . Relations with <100 edges use a global kernel fallback.

B Extended Results

Table 3: Full link prediction metrics on FB15k-237.

Model	MRR	Hits@1	Hits@3	Hits@10
DistMult	0.264	0.184	0.290	0.433
GGPN	0.249	0.169	0.275	0.416
GP-KGE	0.255	0.178	0.280	0.413